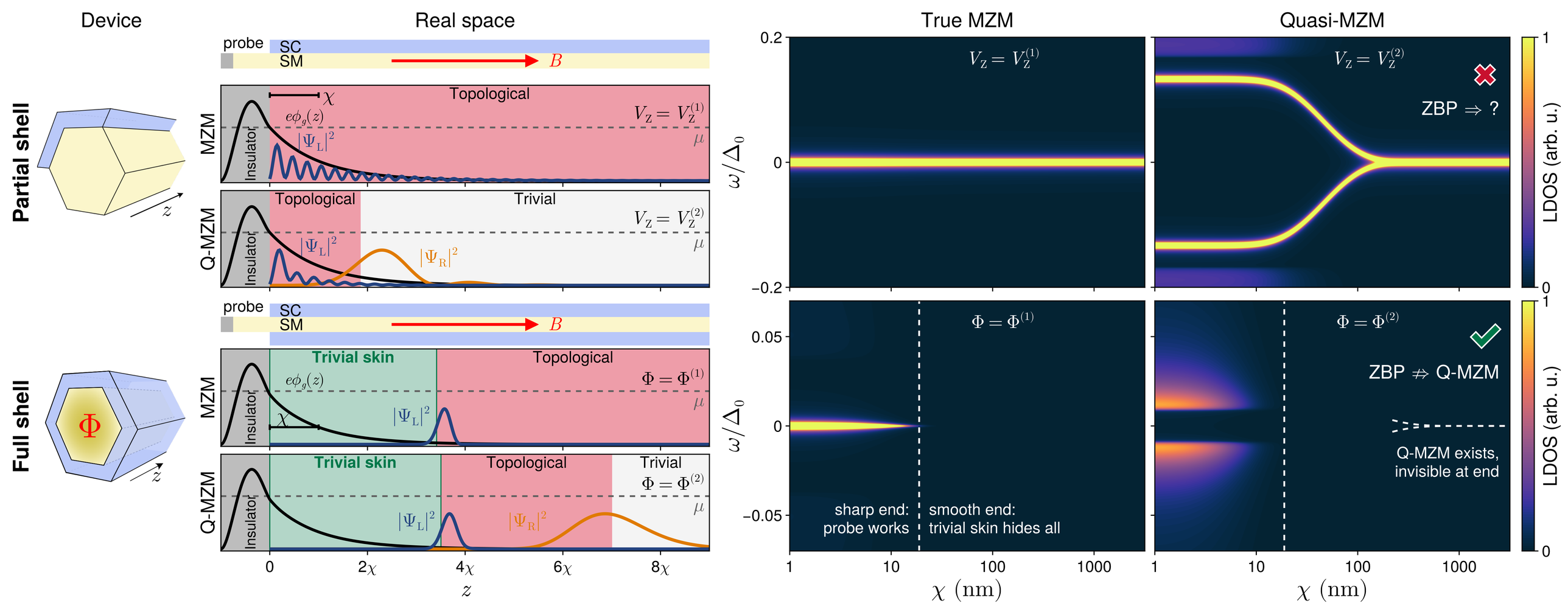

Device
Real space
True MZM
Quasi-MZM
Partial shell
Full shell
probe
SC
SM
B
Topological
Trivial
Trivial skin
MZM
Q-MZM
Insulator
χ
$e\phi_g(z)$
$|\Psi_L|^2$
$|\Psi_R|^2$
μ
$V_Z = V_Z^{(1)}$
$V_Z = V_Z^{(2)}$
$\Phi = \Phi^{(1)}$
$\Phi = \Phi^{(2)}$
Φ
z
0
2χ
4χ
6χ
8χ
ω/Δ_0
0.2
0
−0.2
0.05
−0.05
ZBP ⇒ ?
ZBP ⇏ Q-MZM
Q-MZM exists, invisible at end
sharp end: probe works
smooth end: trivial skin hides all
1
10
100
1000
χ (nm)
LDOS (arb. u.)